MMDU MULLANA OFFICIAL SYLLABUS • BELE-001 • UNIT 3

Unit 3: Semiconductors, Diodes & Transistors (BJT Made Simple)

Subject: Basic Electrical & Electronics Engineering (BELE-001) • **Exam Weightage:** 24 - 28 Marks • **Study Time:** 35 mins

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1. Semiconductors & Doping (The Secret of Silicon)

1. Energy Band Theory & Semiconductor Physics

Solids are classified by the width of the forbidden energy bandgap (E_g) separating the completely filled **Valence Band** from the empty **Conduction Band**:

- **Metals (Conductors):** Valence and conduction bands overlap ($E_g = 0$ eV). Massive sea of free electrons.
- **Insulators (Diamond, Quartz):** Very wide bandgap ($E_g > 5$ eV). Extremely low thermal excitation across gap.
- **Semiconductors (Silicon, Germanium):** Moderate bandgap:

> $$E_g(\text{Si}) \approx 1.12 \text{ eV}, E_g(\text{Ge}) \approx 0.67 \text{ eV} (\text{at } 300 \text{ K})$$

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2. Intrinsic vs Extrinsic Semiconductors & Doping

1. **Intrinsic Semiconductor:** Chemically ultra-pure silicon (99.9999999).

- Thermal excitation generates electron-hole pairs: $n = p = n_i$.
- At room temperature (300 K), $n_i \approx 1.5 \times 10^{10}$ cm^{-3} for Silicon.

2. **Extrinsic (Doped) Semiconductors:**

- **N-Type Semiconductor:** Pure silicon doped with Pentavalent impurity atoms (Phosphorus, Arsenic, Antimony). 5 valence electrons → 4 form covalent bonds, 1 is easily freed into conduction band. Majority carriers = **Electrons** ($n \approx N_D$), Minority carriers = Holes.
- **P-Type Semiconductor:** Pure silicon doped with Trivalent impurity atoms (Boron, Gallium, Indium). 3 valence electrons → creates a vacant bond state (**Hole**). Majority carriers = **Holes** ($p \approx N_A$), Minority carriers = Electrons.

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3. The Law of Mass Action & Fermi-Dirac Statistics

> ✅ **FUNDAMENTAL LAW / RULE:**
> **Mass Action Law:** In any non-degenerate semiconductor at thermal equilibrium, the product of electron and hole concentrations is strictly constant, depending only on temperature and bandgap, regardless of doping:

> $$\mathbf{n \cdot p = n_i^2}$$

Fermi Energy Level (E_F):

The energy level at which the Fermi-Dirac probability of electron occupancy is exactly 50 ($f(E_F) = 0.5$):

- **Intrinsic:** $E_F \approx E_i$ lies virtually at the exact center of the bandgap: $E_i = \frac{E_C + E_V}{2}$.
- **N-Type:** Doping with donors pushes E_F upward toward the conduction band:

$$\mathbf{E_F - E_i = k_B T \ln\left(\frac{N_D}{n_i}\right)}$$

- **P-Type:** Doping with acceptors pulls E_F downward toward the valence band:

$$\mathbf{E_i - E_F = k_B T \ln\left(\frac{N_A}{n_i}\right)}$$

Problem:
In a pure silicon semiconductor at 300 K, the intrinsic carrier concentration is $n_i = 1.5 \times 10^{10}\ \text{cm}^{-3}$. The sample is doped with donor atoms to a concentration of $N_D = 10^{16}\ \text{cm}^{-3}$. Find the majority electron concentration n and minority hole concentration p.

Step 1 🚀: Majority Carrier Concentration
Since $N_D \gg n_i$, virtually all donor atoms are ionized:

$$n \approx N_D = \mathbf{1.0 \times 10^{16}\ \text{cm}^{-3}}$$

Step 2 ⚡: Apply Mass Action Law $n \cdot p = n_i^2$

$$p = \frac{n_i^2}{n} = \frac{(1.5 \times 10^{10})^2}{10^{16}} = \frac{2.25 \times 10^{20}}{10^{16}} = \mathbf{2.25 \times 10^{4}\ \text{cm}^{-3}}$$

🎯 Electrons $n = 10^{16}\ \text{cm}^{-3}$ | Holes $p = 2.25 \times 10^{4}\ \text{cm}^{-3}$ (Drastic suppression of minority carriers).

University Question (8 Marks):
Pure silicon at room temperature has $n_i = 1.5 \times 10^{16}\text{,m}^{-3}$, electron mobility $\mu_n = 0.135\text{,m}^2\text{/V·s}$, and hole mobility $\mu_p = 0.048\text{,m}^2\text{/V·s}$.

1. Find intrinsic conductivity σ_i.

2. Find conductivity after doping with 10^{22} donor atoms/m^3.

Step 1 🚀: Intrinsic Conductivity $\sigma_i = n_i e (\mu_n + \mu_p)$
$\sigma_i = (1.5 \times 10^{16})(1.6 \times 10^{-19})(0.135 + 0.048) = (2.4 \times 10^{-3})(0.183) = 4.39 \times 10^{-4}$,S/m

Step 2 ⚡: Doped N-Type Conductivity $\sigma_n \approx ND\ e\ \mu_n$
$\sigma_n = (10^{22})(1.6 \times 10^{-19})(0.135) = (1600)(0.135) = 216.0$,S/m
Notice that doping by a tiny fraction increased electrical conductivity by **nearly 500,000 times**!

🎯 Intrinsic Conductivity $\sigma_i = 4.39 \times 10^{-4}$ S/m | Doped Conductivity $\sigma_n = 216.0$ S/m.

University Question (10 Marks):
Calculate the displacement of the Fermi energy level EF from the intrinsic level E_i in silicon at T = 300,K doped with 2.0×10^{16},cm^{-3} phosphorus atoms ($n_i = 1.5 \times 10^{10}$,cm^{-3}, kB T = 0.0259,eV).

Step 1 🚀: Fermi Level Offset Equation

$$E_F - E_i = k_B T \ln\left(\frac{N_D}{n_i}\right)$$

Step 2 ⚡: Compute Concentration Ratio
$(ND / n_i) = (2.0 \times 10^{16} / 1.5 \times 10^{10}) = 1.333 \times 10^6$

Step 3 🎯: Evaluate Logarithm and Energy Shift
$\ln(1.333 \times 10^6) = \ln(1.333) + 6\ \ln(10) = 0.2877 + 6(2.3026) = 0.2877 + 13.8155 = 14.103$
EF - E_i = (0.0259,eV) × 14.103 = 0.365,eV
The Fermi level shifts **0.365,eV upward toward the conduction band**.

🎯 Fermi Level Shift: $\mathbf{E_F - E_i = +0.365}$ eV (closer to Conduction Band).

2. PN Junction Diode & The One-Way Turnstile

1. The P-N Junction: Formation, Depletion Region & Electrostatics

When P-type and N-type semiconductors are metallurgically joined in a single crystal:

1. **Diffusion Current:** High hole concentration in P-region causes holes to diffuse into N-region; electrons diffuse from N-region into P-region.

2. **Uncovered Ion Cores:** Diffusing electrons and holes recombine, leaving unneutralized donor ions (N_D^+) on the N-side and unneutralized acceptor ions (N_A^-) on the P-side.

3. **The Depletion Layer (Space Charge Region):** A thin boundary region ($W \approx 0.1 - 1, \mu$m) completely devoid of mobile charge carriers.

4. **Built-In Potential Barrier (V_0 / V_{bi}):** The space charge establishes an internal electric field $\vec{E}$ pointing from N to P that opposes further diffusion:

$$\mathbf{V_0 = V_T \ln\left(\frac{N_A \cdot N_D}{n_i^2}\right)}$$

where $V_T = \frac{k_B T}{q} \approx 26\ \text{mV}$ at $300\ \text{K}$.
- For Silicon: $V_0 \approx 0.7\ \text{Volts}$. For Germanium: $V_0 \approx 0.3\ \text{Volts}$.

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2. Biasing Regimes & The Shockley Diode Equation

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Forward Bias (+ to P, - to N)              Reverse Bias (- to P, + to N)
Vₐpplied opposes V₀                       Vₐpplied aids V₀
Depletion width shrinks: W -> 0            Depletion width widens: W >> W₀
Current surges exponentially:              Current drops to tiny reverse saturation:
ID = Iₛ * [ exp\frac{VD}{eta*VT} - 1 ]     ID = -Iₛ (nano-amperes)
```

Shockley Diode Characteristic Equation:

$$\mathbf{I_D = I_s \left(e^{\frac{V_D}{\eta V_T}} - 1\right)}$$

- I_s: Reverse saturation current (10^{-12} A for Si, 10^{-6} A for Ge).

- η: Ideality factor ($\eta = 1$ for Ge, $\eta \approx 1 - 2$ for Si).

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3. Zener Breakdown vs Avalanche Breakdown & Regulators

When reverse bias voltage reaches critical breakdown threshold V_{BR}:

1. **Zener Breakdown ($V_{BR} < 6\ \text{V}$):** Occurs in **heavily doped junctions** with ultra-thin depletion width ($< 10\ \text{nm}$). High internal field ($> 10^6\ \text{V/cm}$) directly pulls electrons out of covalent bonds via **quantum mechanical tunneling**. Has a **negative temperature coefficient**.

2. **Avalanche Breakdown ($V_{BR} > 6\ \text{V}$):** Occurs in **lightly doped junctions** with wide depletion layers. High electric field accelerates minority carriers to huge kinetic energies, knocking free secondary electrons via **impact ionization** in a chain reaction. Has a **positive temperature coefficient**.

Zener Voltage Regulator Operation:

When biased in the breakdown region, voltage across the Zener stays fixed at V_Z regardless of current variations ($I_{Z,\min} \leq I_Z \leq I_{Z,\max}$):

$$\mathbf{R_s = \frac{V_{in} - V_Z}{I_Z + I_L}}$$

Problem:
A silicon diode with knee voltage $V_0 = 0.7\ \text{V}$ is connected in series with a resistor $R = 220\ \Omega$ across a $5.0\ \text{V}$ DC supply. Find the diode current I_D and voltage across the resistor V_R.

Step 1 🚀: Check Diode State
Input supply $5.0\ \text{V} > 0.7\ \text{V}$, so the diode is **Forward Biased**.

Step 2 ⚡: Resistor Voltage Drop

$$V_R = V_s - V_D = 5.0 - 0.7 = \mathbf{4.3}\ \text{Volts}$$

Step 3 🎯: Diode Current

$$I_D = \frac{V_R}{R} = \frac{4.3\ \text{V}}{220\ \Omega} = \mathbf{0.0195}\ \text{A} = \mathbf{19.55}\ \text{mA}$$

🎯 Resistor Voltage $V_R = 4.3\ \text{V}$ | Forward Current $I_D = 19.55\ \text{mA}$.

University Question (8 Marks):
A 10 V Zener diode is used to regulate a supply voltage that varies from 15 V to 20 V. The load resistance is RL = 1000~\Omega and minimum Zener current is $IZ_{,min}$ = 5 mA. Calculate the maximum value of series resistor R_s.

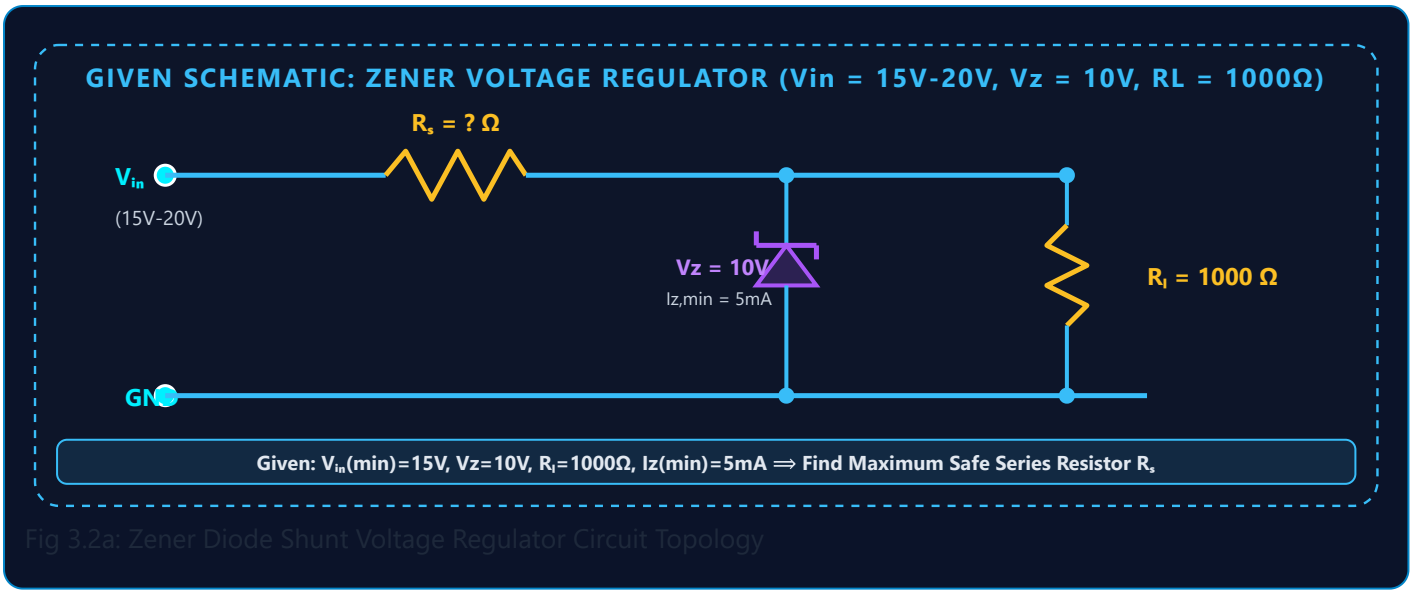


Fig 3.2a: Zener Diode Shunt Voltage Regulator Circuit Topology

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Step 1 🚀: Load Current I_L

$$I_L = \frac{V_Z}{R_L} = \frac{10\ \text{V}}{1000\ \Omega} = 10.0\ \text{mA}$$

Step 2 ⚡: Minimum Input Voltage Condition ($V_{in,\min} = 15\ \text{V}$)

To guarantee regulation at minimum input, the Zener must still carry at least $I_{Z,\min} = 5.0\ \text{mA}$.

Total current: $I_s = I_{Z,\min} + I_L = 5.0 + 10.0 = 15.0\ \text{mA}$.

Step 3 🎯: Determine Maximum Series Resistor R_s

$R_{s,\max} = \frac{V_{in,\min} - V_Z}{I_s} = \frac{15-10}{15\times10^{-3}} = \frac{5}{0.015} = \mathbf{333.3\ \Omega}$

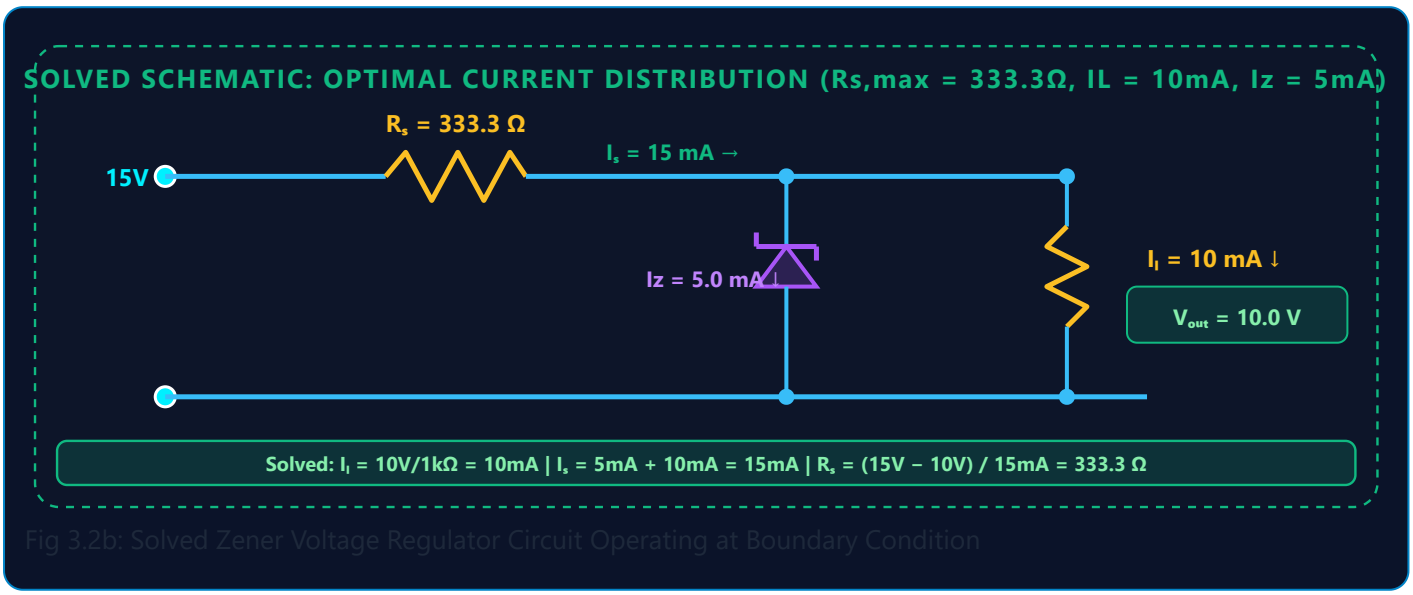


Fig 3.2b: Solved Zener Voltage Regulator Circuit Operating at Boundary Condition

🎯 Maximum Series Resistor $R_{s,\max} = 333.3\ \Omega$.

University Question (10 Marks):

A silicon diode has reverse saturation current $I_s = 10^{-12}$,A and ideality factor $\eta = 1$.

1. Calculate the diode current at a forward bias of VD = 0.65 V at T = 300,K (VT = 26 mV).

2. Determine the dynamic (AC) resistance rd at this operating point.

Step 1 🚀: Apply Shockley Equation

$$I_D = I_s\,(e^{V_D/(etaV_T)} - 1)$$

$$\frac{V_D}{V_T} = \frac{0.65}{0.026} = 25.0$$

$$I_D = 10^{-12}\,(e^{25} - 1) = 10^{-12}(7.20 \times 10^{10}) = 0.0720\ \text{A} = 72.0\ \text{mA}$$

Step 2 ⚡: Calculate Dynamic Resistance rd

rd = (η VT / ID) = (0.026 V / 0.072 A) = 0.361~\Omega

🎯 Forward Current $I_D = 72.0\ \text{mA}$ | Dynamic Resistance $r_d = 0.361\ \Omega$.

3. Rectifiers, Clippers & Clampers (AC to Clean DC)

1. Rectification Circuits & Performance Analysis

A **rectifier** converts bidirectional alternating voltage (AC) into pulsating unidirectional direct voltage (DC).

Performance Metrics Catalog:

- **Rectification Efficiency (η):** $\eta = \frac{P_{dc}}{P_{ac}} = \frac{I_{dc}^2 R_L}{I_{rms}^2 (R_f + R_L)}$.
- **Ripple Factor (γ):** Ratio of RMS AC component to DC component:

$$\gamma = \frac{\mathbf{V_{ac,rms}}}{\mathbf{V_{dc}}} = \sqrt{\left(\frac{\mathbf{V_{rms}}}{\mathbf{V_{dc}}}\right)^2 - \mathbf{1}}$$

- **Peak Inverse Voltage (PIV):** Maximum reverse voltage across a non-conducting diode.

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2. Mathematical Comparison of the Three Standard Rectifier Topologies

Metric	Half-Wave Rectifier	Center-Tapped Full-Wave	Full-Wave Bridge Rectifier
Diode Count	\frac{1}{2}	**4**	
Center-Tap Transformer Needed?	No	**Yes (Expensive, Bulky)**	**No (Standard transformer)**
Peak Inverse Voltage (PIV)	V_m	$\mathbf{2V_m}$	$\mathbf{V_m}$ *(Economical!)*
DC Output Voltage (V_{dc})	$\frac{V_m}{\pi} \approx 0.318 V_m$	$\frac{2V_m}{\pi} \approx 0.637 V_m$	$\frac{\mathbf{2V_m}}{\pi} \approx \mathbf{0.637V_m}$
RMS Output Voltage (V_{rms})	$\frac{V_m}{2} = 0.50 V_m$	$\frac{V_m}{\sqrt{2}} \approx 0.707 V_m$	$\frac{\mathbf{V_m}}{\sqrt{\mathbf{2}}} \approx \mathbf{0.707V_m}$
Max Efficiency ($\eta_{\max}$)	**40.6%**	**81.2%**	**81.2%**
Ripple Factor (γ)	**1.21**	**0.482**	**0.482**
Fundamental Ripple Frequency	f (50 Hz)	$2f$ (100 Hz)	**2f** (100 Hz, easier to filter!)

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3. Waveform Shapers: Clippers, Clampers & Filters

1. **Clippers (Limiters):** Non-linear wave-shaping networks that remove a portion of an input signal above or below a specified reference voltage without distorting the remaining waveform.

2. **Clampers (DC Restorers):** Networks comprising a diode, resistor, and capacitor that clamp either the positive or negative peak of an input waveform to a specified DC reference level, preserving the original peak-to-peak amplitude $2V_m$.

3. **Capacitor Filter (C-Filter):** A large capacitor placed in parallel with load resistor R_L charges to peak V_m and slowly discharges through R_L with time constant $\tau = R_L C \gg T$:

$$\mathbf{V_{r(p-p)} = \frac{I_{dc}}{2fC}}, \gamma = \frac{1}{4\sqrt{3}\mathbf{fCR_L}}$$

Problem:
A full-wave bridge rectifier is fed from a 230 V/12 V transformer. Assuming ideal diodes, calculate the DC output voltage V_{dc} and Peak Inverse Voltage (PIV) across each diode.

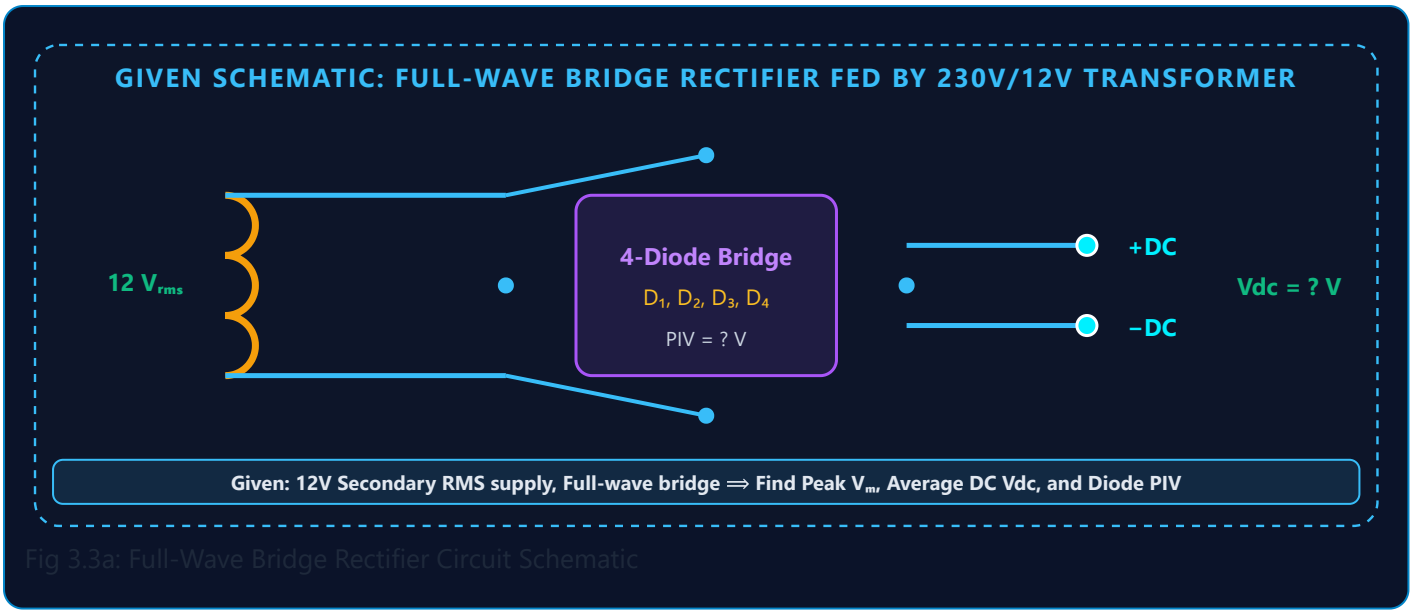


Fig 3.3a: Full-Wave Bridge Rectifier Circuit Schematic

Step 1 🚀: Secondary Peak Voltage V_m

$$V_{rms} = 12.0 \text{ V} \Rightarrow V_m = \sqrt{2} \times 12.0 = \mathbf{16.97} \text{ Volts}$$

Step 2 ⚡: DC Output Voltage V_{dc}

$$V_{dc} = \frac{2V_m}{\pi} = \frac{2 \times 16.97}{\pi} = \mathbf{10.80} \text{ Volts}$$

Step 3 🎯: Peak Inverse Voltage (PIV)

For a bridge rectifier:

PIV $= V_m = \mathbf{16.97}$ Volts

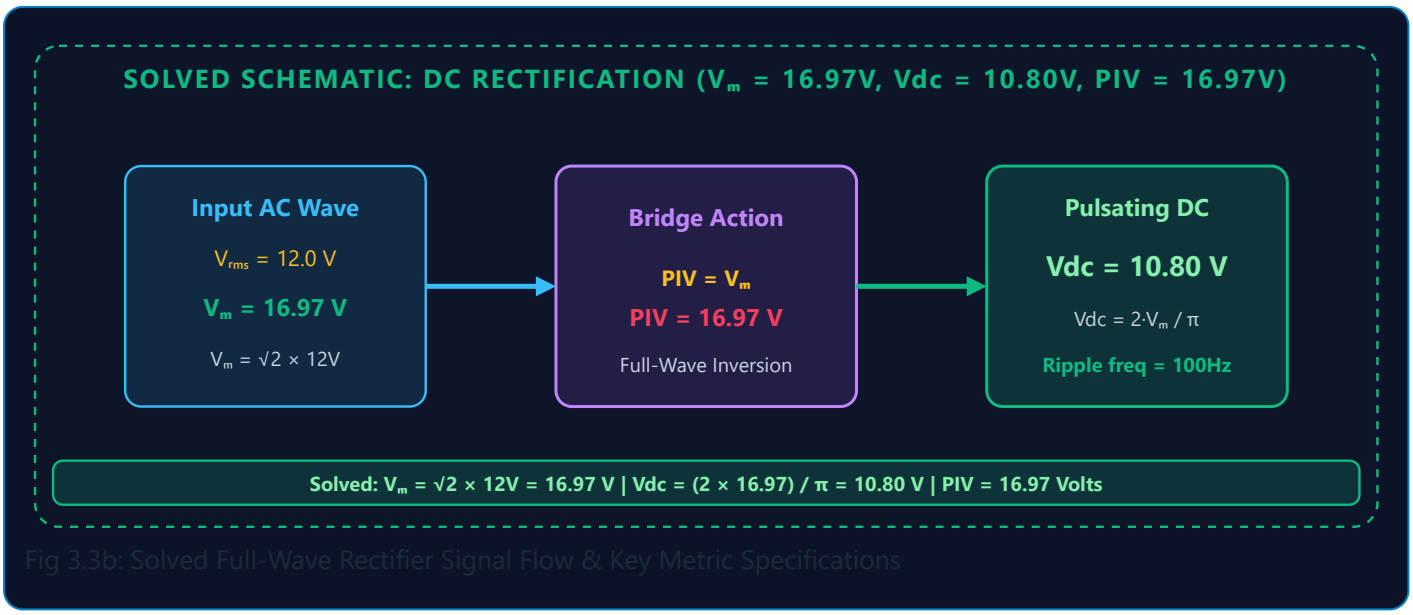


Fig 3.3b: Solved Full-Wave Rectifier Signal Flow & Key Metric Specifications

🎯 $V_{dc} = 10.80$ V | Diode PIV rating needed = 16.97 V.

University Question (8 Marks):
A full-wave rectifier delivers power to a 500~\Omega load resistor from a secondary voltage of v(t) = 50 \sin(314t). Each diode has forward resistance Rf = 20~\Omega.
Calculate: (1) DC load current Idc, (2) Rectification efficiency η, (3) Ripple factor γ.

Step 1 🚀: DC Load Current
I_m = (V_m / Rf + RL) = \frac{50}{20 + 500} = \frac{50}{520} = 0.09615 A
Idc = (2 I_m / π) = (2 × 0.09615 / π) = 0.0612 A = 61.2,\text(mA)

Step 2 ⚡: Rectification Efficiency η
η = (Pdc / P_ac) = (0.812 / 1 + (Rf / RL)) = (0.812 / 1 + \frac{20}{500}) = (0.812 / 1.04) = 78.08%

Step 3 🎯: Ripple Factor

$$\gamma = \sqrt{\left(\frac{I_{\text{rms}}}{I_{\text{dc}}}\right)^2 - 1} = \sqrt{(1.11)^2 - 1} = 0.482$$

🎯 $I_{dc} = 61.2\ \text{mA}$ | Efficiency $\eta = 78.08$ | Ripple Factor $\gamma = 0.482$.

University Question (10 Marks):
A full-wave bridge rectifier is supplied by a 50 Hz AC source with peak voltage V_m = 24 V. The DC load draws Idc = 200 mA.
Design a shunt capacitor filter such that the peak-to-peak ripple voltage V_r does not exceed 1.0 V. Also determine the resulting ripple factor γ.

Step 1 🚀: Capacitor Filter Sizing Formula
For full-wave rectifier, ripple frequency is 2f = 2 × 50 = 100 Hz.
$V_{r(p-p)}$ = \frac{Idc}{2 f C}
1.0 = (0.20 / 2(50) C) = (0.20 / 100 C) ⇒ C = (0.20 / 100) = 2.0 × 10^{-3},F = ^{2000},μF

Step 2 ⚡: DC Output Voltage with Filter
Vdc = V_m - ($V_{r(p-p)}$ / 2) = 24 - (1.0 / 2) = 23.5,Volts

Step 3 🎯: Ripple Factor γ
γ = ($V_{r(rms)}$ / Vdc) = ($V_{r(p-p)}$ / 2√(3) · Vdc) = (1.0 / 2√(3) × 23.5) = (1.0 / 81.41) = 0.0123 = 1.23%

🎯 Filter Capacitor $C = 2000, \mu\text{F}$ | Output $V_{dc} = 23.5\ \text{V}$ | Ripple Factor $\gamma = 1.23$ (Ultra-smooth DC).

4. Bipolar Junction Transistor (BJT) — The Valve Analogy

1. Bipolar Junction Transistors (BJT): Physical Structure & Working

A **Bipolar Junction Transistor** is a three-terminal, two-junction semiconductor device whose operation relies on transport of both majority and minority charge carriers (bipolar).

The Three Structural Regions:

1. **Emitter (E):** Heavily doped to inject maximum majority carriers into the base. Moderate physical size.
2. **Base (B):** Ultra-thin ($< 1, \mu\text{m}$) and extremely lightly doped to minimize electron-hole recombination (< 2).
3. **Collector (C):** Moderately doped, physically the largest region to safely dissipate Joule heat generated by collecting carriers.

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2. The Three Operating Regimes & Biasing Configurations

Operating Region	[Emitter-Base Junction (EBJ)]	[Collector-Base Junction (CBJ)]	[Circuit Application]
Active Region	**Forward Biased**	**Reverse Biased**	Linear Analog Amplifier
Saturation Region	**Forward Biased**	**Forward Biased**	Digital Switch ON (Closed) ($V_{CE\,\text{sat}} \approx 0.2\ \text{V}$)
Cutoff Region	**Reverse Biased**	**Reverse Biased**	Digital Switch OFF (Open) ($I_C \approx 0$)
Inverse Active	Reverse Biased	Forward Biased	Attenuator (Rarely used)

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3. Current Gain Relations (α, β, γ) & Characteristics

1. **Current Continuity Equation:**

$$\mathbf{I_E = I_B + I_C}$$

2. **Common-Base Current Gain (α):**

$$\alpha = \frac{\mathbf{I_C}}{\mathbf{I_E}} \approx \mathbf{0.95 - 0.998}$$

3. **Common-Emitter Current Gain (β or h_{FE}):**

$$\beta = \frac{\mathbf{I_C}}{\mathbf{I_B}} \approx \mathbf{50 - 300}$$

4. Mathematical Derivation of β in terms of α:

$$I_E = I_B + I_C \Rightarrow \frac{I_E}{I_C} = \frac{I_B}{I_C} + 1 \Rightarrow \frac{1}{\alpha} = \frac{1}{\beta} + 1$$

$$\frac{1}{\beta} = \frac{1}{\alpha} - 1 = \frac{1-\alpha}{\alpha} \Rightarrow \beta = \frac{\alpha}{\mathbf{1}-\alpha}, \alpha = \frac{\beta}{\beta+\mathbf{1}}$$

DC Operating Point (Q-Point) and Load Line:

Applying KVL to the collector-emitter loop of a Common-Emitter amplifier:

$$\mathbf{V_{CE} = V_{CC} - I_C R_C}$$

- Upper intercept (Saturation limit): $V_{CE} = 0 \Rightarrow I_{C\,\text{sat}} = \frac{V_{CC}}{R_C}$.
- Lower intercept (Cutoff limit): $I_C = 0 \Rightarrow V_{CE\,\text{cutoff}} = V_{CC}$.

Problem:
A transistor has a base current $I_B = 50, \mu\text{A}$ and collector current $I_C = 4.95\ \text{mA}$. Find: (1) Emitter current I_E, (2) Current gain β, (3) Current gain α.

Step 1 🚀: Emitter Current $I_E = I_C + I_B$

$$I_E = 4.95\ \text{mA} + 0.050\ \text{mA} = \mathbf{5.0}\ \text{mA}$$

Step 2 ⚡: Current Gain β

$$\beta = \frac{I_C}{I_B} = \frac{4.95 \times 10^{-3}}{50 \times 10^{-6}} = \mathbf{99.0}$$

Step 3 🎯: Current Gain α

$$\alpha = \frac{\beta}{\beta + 1} = \frac{99}{100} = \mathbf{0.99}$$

🎯 $I_E = 5.0$ mA | Common-Emitter Gain $\beta = 99$ | Common-Base Gain $\alpha = 0.99$.

University Question (8 Marks):

In a Common-Emitter BJT circuit, VCC = 12 V, RC = 2.0,k\Omega, RB = 200,k\Omega, and β = 100. Assuming VBE = 0.7 V, find:

1. Base current IB.

2. Operating point (IC, VCE).

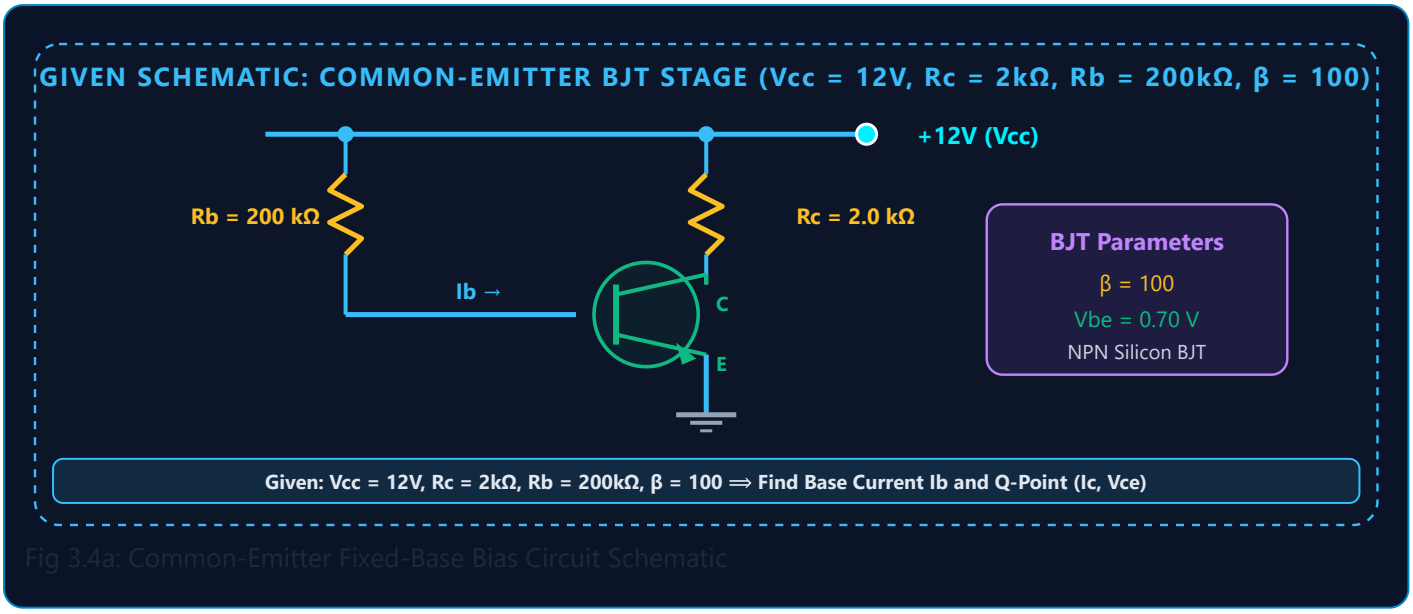


Fig 3.4a: Common-Emitter Fixed-Base Bias Circuit Schematic

Step 1 🚀: Base Current I_B

$$I_B = \frac{V_{CC} - V_{BE}}{R_B} = \frac{12 - 0.7}{200 \times 10^3} = \frac{11.3}{200{,}000} = \mathbf{56.5}, \mu\text{A}$$

Step 2 ⚡: Collector Current I_C

$$I_C = \beta I_B = 100 \times (56.5 \times 10^{-6}) = \mathbf{5.65}\ \text{mA}$$

Step 3 🎯: Collector-Emitter Voltage V_{CE}

$V_{CE} = V_{CC} - I_C R_C = 12 - (5.65 \times 10^{-3})(2000) = 12 - 11.3 = \mathbf{0.70}$ Volts

The transistor is in the active region near the edge of saturation!

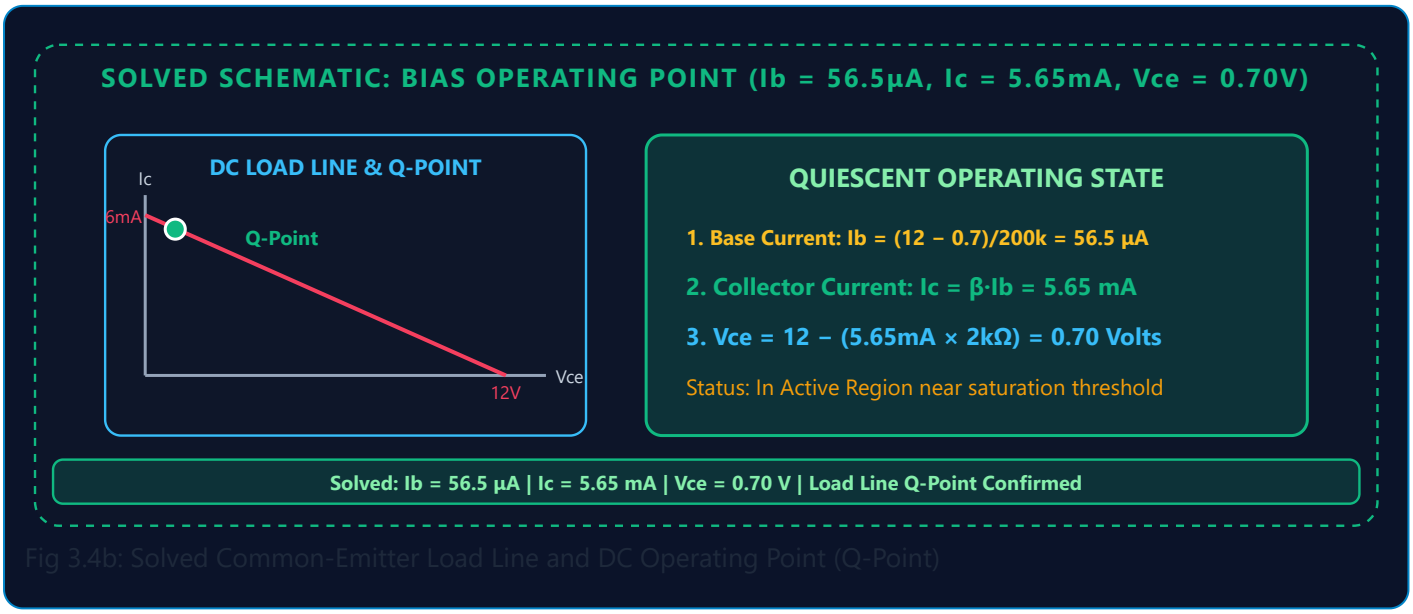


Fig 3.4b: Solved Common-Emitter Load Line and DC Operating Point (Q-Point)

🎯 Q-Point: $\mathbf{I_C = 5.65\ mA}$,; $\mathbf{V_{CE} = 0.70\ V}$.

University Question (12 Marks - University Exam):
Design a voltage-divider biased Common-Emitter amplifier stage with VCC = 15 V, quiescent point IC = 2.0 mA, VCE = 7.5 V, and stability factor condition VE = 0.1 VCC. Transistor has β = 100, VBE = 0.7 V. Find RE, RC, R_1, R_2.

Step 1 🚀: Determine RE and RC
VE = 0.1 VCC = 0.1(15) = 1.5 V
RE = (VE / IE) ≈ (VE / IC) = (1.5 V / 2.0 mA) = 750~\Omega
From collector loop: VCC = VCE + IC RC + VE:
15 = 7.5 + (2.0×10^{-3}) RC + 1.5 ⇒ 2.0×10^{-3} RC = 6.0 ⇒ RC = 3.0,k\Omega

Step 2 ⚡: Base Voltage VB
VB = VE + VBE = 1.5 + 0.7 = 2.2,Volts

Step 3 🎯: Size Bias Resistors R_1 and R_2
For excellent thermal stability, make divider current I_1 ≈ 10 IB:
IB = (IC / β) = (2.0 mA / 100) = 20 µA ⇒ Id_{iv} = 10 IB = 200 µA
R_2 = (VB / Id_{iv}) = (2.2 V / 200 µA) = 11.0,k\Omega
R_1 = (VCC - VB / Id_{iv}) = (15 - 2.2 / 200 µA) = (12.8 / 200 µA) = 64.0,k\Omega

🎯 Designed Bias Stage: $R_C = 3.0\ \text{k}\Omega, R_E = 750\ \Omega, R_1 = 64\ \text{k}\Omega, R_2 = 11\ \text{k}\Omega$.

University Solved Examination Questions (Unit 3)

MMDU Official University Solved Examination Questions (Unit 3)

Course: Basic Electrical & Electronics Engineering (`BELE-001`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

📘 Question 7 (1 Mark): Barrier Potential of Silicon vs. Germanium

Question:
Why is the forward knee voltage of Silicon (0.7V) higher than that of Germanium (0.3V)?

> 📊 **Question Schematic / State Specification:**

```
Question Bandgap Atomic Dimension: Si (Eg = 1.1 eV, tightly bound valence electrons) vs Ge (Eg = 0.67 eV).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Barrier Height Equation:

        `V₀ = VT \ln(NA ND / nᵢ²)`. Higher bandgap in Si yields lower intrinsic carrier density `nᵢ` ⟹ higher built-in potential `V₀`.
```

Silicon's larger energy bandgap (1.1 eV vs 0.67 eV) creates a stronger internal electric field across the depletion region, requiring 0.7V of forward bias to overcome compared to 0.3V for Germanium.

🎯 **Verified Result:** Question 7 step-by-step resolution per MMDU university marking scheme.

📘 Question 8 (1 Mark): Peak Inverse Voltage in Bridge vs. Center-Tapped Rectifier

Question:
Compare the Peak Inverse Voltage (PIV) rating demanded of diodes in Bridge vs Center-Tapped rectifiers.

> 📊 **Question Schematic / State Specification:**

```
Question Reverse Voltage Stress: Bridge Diode sees Vₘ | Center-Tapped Diode sees 2 Vₘ.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution PIV Ratio Formulation:

        `PIVBridge = Vₘ` versus `PIVCenter-Tap = 2 Vₘ`.
```

In a center-tapped full-wave rectifier, each non-conducting diode must withstand the full secondary voltage $2\ V_m$. In a bridge rectifier, the reverse voltage divides across two reverse diodes, requiring only V_m rating.

🎯 **Verified Result:** Question 8 step-by-step resolution per MMDU university marking scheme.

📘 Question 9 (1 Mark): BJT Base Width Design Principle

Question:
Why is the base region of a BJT manufactured ultra-thin and lightly doped?

> 📊 **Question Schematic / State Specification:**

```
Question Carrier Diffusion Path: Emitter (Heavy Injection) ➔ Base (Minimal Recombination) ➔ Collector (Extraction).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Recombination Minimization Model:

        `Base Recombination &amp;lt; 2% ⇒ IC ≈ IE ⇒ α = IC/IE ≈ 0.98–0.99`.
```

A thin, lightly doped base ensures that almost all injected minority carriers traverse the base without recombining, allowing over 98% of carriers to be swept into the collector for high current gain β.

🎯 **Verified Result:** Question 9 step-by-step resolution per MMDU university marking scheme.

📘 Question 15 (2 Marks): Zener Shunt Regulator Load Current Calculation

Question:

A 10V Zener regulator has input V_{in} = 20V, series resistor RS = 200Ω, and load RL = 1 kΩ. Calculate source current IS, load current IL, and Zener current IZ.

> 📊 **Question Schematic / State Specification:**

```
Question Shunt Circuit Diagram: 20V DC ➔ RS (200Ω) ➔ Node VZ (10V) ➔ Parallel branches: Zener Diode & RL (1kΩ).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Current Distribution Formulation:

        `IS = (Vin - VZ) / RS = (20 - 10) / 200 = 10 / 200 = 0.05  A = 50  mA`

        `IL = VZ / RL = 10  V / 1000  Ω = 0.01  A = 10  mA`

        `IZ = IS - IL = 50  mA - 10  mA = 40  mA`
```

The Zener diode safely conducts 40 mA, maintaining a steady 10V regulated output across the 1 kΩ load.

🎯 **Verified Result:** Question 15 step-by-step resolution per MMDU university marking scheme.

📘 Question 19 (4 Marks): Derivation of Full-Wave Bridge Rectifier Ripple Factor

Question:

Derive the mathematical expression for Ripple Factor γ of a Full-Wave Bridge Rectifier and prove it equals 0.482.

> 📊 **Question Schematic / State Specification:**

```
Question Rectified Waveform: Pulsating DC with period T/2: v(t) = |Vₘ sin(ωt)|.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Ripple Factor Derivation Trace:

        `Ripple Factor &gamma; = Vₐc,ᵣₘₛ / Vdc = √{(Vᵣₘₛ/Vdc)² - 1}`

        `Vdc = \frac{1}{π} ∫_0^π Vₘ \sin(θ) dθ = (2 Vₘ) / π ≈ 0.6366 Vₘ`

        `Vᵣₘₛ = √{ \frac{1}{π} ∫_0^π Vₘ² \sin²(θ) dθ } = Vₘ / √2 ≈ 0.7071 Vₘ`

        `Form Factor Kf = Vᵣₘₛ / Vdc = \frac{Vₘ / √2}{2 Vₘ / π} = π / (2 √2) ≈ 1.1107`

        `&gamma; = √{(1.1107)² - 1} = √(1.2337 - 1) = √(0.2337) = 0.4834 ≈ 0.482`
```

The ripple factor evaluates to 0.482, indicating that AC harmonic content is less than half of the DC component without filtering.

🎯 **Verified Result:** Question 19 step-by-step resolution per MMDU university marking scheme.

📘 Question 23 (6 Marks): BJT Common Emitter Input and Output Characteristics

Question:
Sketch and thoroughly explain the Input and Output static characteristics of an NPN Bipolar Junction Transistor in Common Emitter (CE) configuration. Define Early Effect (Base-Width Modulation) and explain how the BJT acts as a digital switch.

> 📊 **Question Schematic / State Specification:**

```
Question Characteristic Plots: Input (IB vs VBE at constant VCE) | Output (IC vs VCE at constant IB showing Cutoff, Active, and Saturation).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution 3 Operating Regimes & Early Effect:

        `1. Cutoff (VBE ≤ 0, VCE = VCC): Both junctions reverse biased; IC ≈ 0 ⇒ Switch OFF (Open)`

        `2. Saturation (VBE &gt; 0.7V, VCE ≈ 0.2V): Both junctions forward biased; IC = IC,sat ⇒ Switch ON (Closed)`

        `3. Active (VBE &gt; 0.7V, VCE &gt; 1V): EBJ forward, CBJ reverse; linear relation IC = β IB ⇒ Linear Amplifier`

        `Early Effect: As VCE &uarr; ⟹ Reverse bias on CBJ &uarr; ⟹ Depletion width &uarr; ⟹ Effective base width Wb &darr; ⟹ IC slopes upward`
```

The Early Effect causes collector current curves to have a slight positive slope in the active region; projecting these curves backward intercepts the negative voltage axis at the Early Voltage `-VA`.

🎯 **Verified Result:** Question 23 step-by-step resolution per MMDU university marking scheme.
